American Options and Optimal Stopping

When exercising is a decision, not just a payoff

Lecture 10

One new right changes the pricing problem

The question

An American option lets its holder choose *when* to exercise. How do we put a no-arbitrage value on a payoff whose exercise date is itself a decision?

1. Separate the right to exercise from the payoff received on exercise.
2. Turn the exercise date into a stopping-time problem.
3. Solve the discrete problem by backward induction.
4. Read the continuous-time version as a free-boundary problem.
5. See how trees, PDEs, and simulation implement the idea.

The exercise right

European and American options have the same payoff map

Let $g(S) = (S - K)^+$ for a call or $g(S) = (K - S)^+$ for a put.

European option

Exercise is fixed at maturity:

$$\text{payoff} = g(S_T).$$

There is no exercise decision before T.

American option

The holder chooses an admissible time

$$t \le \tau \le T, \qquad \text{payoff} = g(S_\tau).$$

The payoff map is unchanged; the timing right is valuable.

NO-ARBITRAGE Maturity is always available. Therefore an American option cannot be worth less than its European counterpart.

The first inequality is immediate

For the same strike, underlying, and maturity,

$$\boxed{V_t^{\mathrm{Am}} \geq V_t^{\mathrm{Eu}}.}$$

Why?

The American holder can choose $\tau = T$, thereby reproducing the European payoff. Any better exercise rule can only add value.

YOUR TURN Does *in the money* automatically mean *exercise now* for an American option?

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No

An in-the-money option should be exercised only if $g(S_t)$ is at least its *continuation value*.

Early exercise has an economic cost and benefit

Exercise a put now

Receive $K - S_t$ and invest the cash. This can be attractive when the put is deep in the money and interest rates are positive.

Wait instead

Keep downside insurance and optionality. A further fall in the stock can increase the eventual payoff.

WATCH OUT An American call on a non-dividend-paying stock is a special case: under the usual Black–Scholes assumptions, early exercise is never optimal. A call on a dividend-paying stock may be different.

A stopping time prevents clairvoyance

At date t, an exercise rule may use only information known at t. Formally, τ is a stopping time if

$$\{\tau = t\} \in \mathcal{F}_t \qquad \text{for every } t.$$

Admissible rule

Exercise the put the first time the stock is below 80.

Not admissible

Exercise today if tomorrow's stock will be below 80.

NO-ARBITRAGE The word *stopping* encodes the information constraint. It makes the exercise choice economically meaningful.

Discrete time: value by backward induction

The American price is an optimal-stopping value

Work on dates $0, 1, \ldots, T$, with one-period gross rate $R = 1 + r$ and discount factor $\beta = R^{-1}$. Under a risk-neutral measure $\mathbb{Q}$,

$$V_t = \operatorname*{ess\,sup}_{\tau \in \mathcal{T}_t} \mathbb{E}^{\mathbb{Q}}\left[\beta^{\tau - t} g(S_\tau) \mid \mathcal{F}_t\right],$$

where $\mathcal{T}_t$ contains stopping times in $\{t, \ldots, T\}$.

Interpretation

At each node, evaluate every feasible future exercise rule under $\mathbb{Q}$, then choose the best one. The rule may adapt to information revealed along the path.

At every node there are exactly two actions

Exercise

Stop now and receive intrinsic value

$$\boxed{g(S_t).}$$

No future uncertainty remains.

Continue

Keep the option for one period:

$$\boxed{C_t = \beta \mathbb{E}^{\mathbb{Q}}[V_{t+1} \mid \mathcal{F}_t].}$$

Tomorrow's value already reflects later optimal choices.

NO-ARBITRAGE Backward induction gives

$$\boxed{V_T = g(S_T), \qquad V_t = \max\{g(S_t), C_t\}.}$$

The exercise rule follows the comparison

$$\text{exercise at } t \quad \Longleftrightarrow \quad g(S_t) \geq \beta \mathbb{E}^{\mathbb{Q}}[V_{t+1} \mid \mathcal{F}_t].$$

	Intrinsic $g(S_t)$	Continuation C_t
Exercise region	$g(S_t) \geq C_t$	stop
Continuation region	$g(S_t) < C_t$	wait

Worked example: a two-period American put

MODEL

Initial stock	$S_0 = 100$
Up/down factors	$u = 1.2, \quad d = 0.8$
Gross risk-free return	$R = 1.05$
Strike	$K = 105$

The risk-neutral up probability is

$$p = \frac{R - d}{u - d} = \frac{1.05 - 0.8}{1.2 - 0.8} = 0.625, \qquad 1 - p = 0.375.$$

At date 1, stock values are $120, 80$. At maturity, the recombining tree has values $144, 96, 64$.

Step 1: write the terminal payoffs

For the put $g(S) = (105 - S)^+$:

S_2	144	96	64
$g(S_2)$	0	9	41

The node with $S_1 = 120$

Immediate exercise gives 0. Continuing gives

$$C_1^{(120)} = \frac{0.625(0) + 0.375(9)}{1.05} = 3.21.$$

So we continue, and $V_1^{(120)} = 3.21$.

Step 2: is early exercise optimal at $S_1 = 80$?

YOUR TURN At $S_1 = 80$, the immediate put payoff is $105 - 80 = 25$. The two terminal payoffs after waiting are 9 and 41. Should the holder exercise or continue?

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Compare the two values

$$g(80) = 25, \qquad C_1^{(80)} = \frac{0.625(9) + 0.375(41)}{1.05} = 20.$$

Since $25 > 20$, exercise at this node and set $V_1^{(80)} = 25$.

NO-ARBITRAGE The optimal action depends on the node. The American put has an exercise region, not a pre-announced exercise date.

Step 3: return to time zero

At $t = 0$, immediate exercise is worth $g(100) = 5$. Continuing is worth

$$C_0 = \frac{0.625(3.21) + 0.375(25)}{1.05} = 10.84.$$

Therefore

$$\boxed{V_0^{\mathrm{Am}} = \max\{5, 10.84\} = 10.84.}$$

The policy

Wait at time 0; at time 1, exercise only in the down node $S_1 = 80$. The price and the policy are computed together.

The tree records both the exercise policy and the hedge

Date	Stock	Exercise value	Continuation value	Decision
0	100	5	10.84	continue
1, up	120	0	3.21	continue
1, down	80	25	20	exercise

At time 0, the one-period replicating hedge for the two future option values 3.21 and 25 uses

$$\Delta_0 = \frac{3.214 - 25}{120 - 80} = -0.545, \qquad B_0 \approx 65.31.$$

NO-ARBITRAGE $\Delta_0 S_0 + B_0 = 10.84$. The hedge finances the option while it continues; at the down node it instead finances the exercise payment 25.

The early-exercise premium is visible in the numbers

If exercise were allowed only at maturity, the European put price would be

$$V_0^{\mathrm{Eu}} = \frac{2(0.625)(0.375)(9) + (0.375)^2(41)}{1.05^2} = 9.06.$$

Hence

$$\boxed{V_0^{\mathrm{Am}} - V_0^{\mathrm{Eu}} = 10.84 - 9.06 = 1.78.}$$

What creates the difference?

Not a different terminal payoff. It is the right to take 25 at the down node rather than be forced to carry the position to maturity.

A fast decision check

YOUR TURN At a node of an American put, intrinsic value is 18 and discounted risk-neutral continuation value is 21. What is the action and what is the option value at the node?

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Answer

Continue. The option is worth $\max\{18, 21\} = 21$. Being in the money is not enough: exercising would throw away 3 of continuation value.

The Snell envelope is the discounted value process

Define discounted exercise payoff and value by

$$Y_t = \beta^t g(S_t), \qquad \widetilde{V}_t = \beta^t V_t.$$

The recursion becomes

$$\widetilde{V}_T = Y_T, \qquad \widetilde{V}_t = \max\bigl\{ Y_t, \mathbb{E}^{\mathbb{Q}}[\widetilde{V}_{t+1} \mid \mathcal{F}_t] \bigr\}.$$

Snell envelope

$\widetilde{V}$ is the smallest $\mathbb{Q}$-supermartingale that dominates the discounted payoff process Y.

NO-ARBITRAGE It is the American analogue of risk-neutral valuation: the minimal discounted process that can always cover an exercise request. The next slide shows this is *both* the buyer's optimal-stopping value and the seller's minimal hedging cost — the same number, seen from two sides.

The seller must superhedge every admissible exercise date

Buyer's value

The recursion *is* the buyer's best achievable outcome:

$$V_t = \max\{g(S_t), \beta\mathbb{E}^{\mathbb{Q}}[V_{t+1} \mid \mathcal{F}_t]\}.$$

No stopping rule can do better for the holder.

Seller's price

The seller does not know which τ the buyer will pick. A self-financing X^π must satisfy

$$X_u^\pi \geq g(S_u) \qquad \text{for every admissible } u \geq t.$$

The no-arbitrage price is the cheapest such protection:

$$\boxed{V_t^{\mathrm{Am}} = \inf\{x : \exists\, \pi \text{ self-financing with } X_u^\pi \geq g(S_u) \text{ for every } u \geq t\}.}$$

NO-ARBITRAGE Tracking V_t with a replicating trade already superhedges every node, so it costs no more than V_t; no cheaper strategy survives every stopping choice the buyer might make. Seller's minimal capital = buyer's optimal value.

Continuous time: an obstacle and a boundary

In Black–Scholes, the choice is over all future times

Under $\mathbb{Q}$, suppose

$$\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^{\mathbb{Q}}.$$

Why work directly under $\mathbb{Q}$?

Black–Scholes pricing uses discounted expectations under the risk-neutral measure $\mathbb{Q}$ (Lectures 6–8). The discrete recursion already used this measure; we now let exercise range over all stopping times.

For an American payoff g, its value function is

$$\boxed{v(t,s) = \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}}\left[e^{-r(\tau-t)} g(S_\tau) \mid S_t = s\right].}$$

Exercise region

$v(t,s) = g(s)$: take the payoff now.

Continuation region

$v(t,s) > g(s)$: keep the option alive.

From an exercise grid to the continuous-time PDE

First restrict exercise to $0, h, 2h, \ldots, T$, and denote this **Bermudan** value by v_h. At a grid date $t < T$,

$$\boxed{v_h(t,s) = \max\left\{g(s),\ \mathbb{E}^{\mathbb{Q}}_{t,s}\left[e^{-rh} v_h(t+h, S_{t+h})\right]\right\}.}$$

Let the exercise mesh tend to zero

As $h \downarrow 0$, the grid values approach the American value. In the continuation region, a short-time expansion yields the Black–Scholes PDE; the stopping branch gives $v \geq g$.

WATCH OUT For a finite h, this recursion excludes exercise inside $(t, t+h)$. It is not an exact finite-step identity for continuous-time exercise.

The PDE has an obstacle

Exercise branch

Stopping attains the max: $v(t,s) = g(s)$. An American value must stay above immediate exercise, $v(t,s) \geq g(s)$.

Continuation branch

Continuing attains the max: expanding $\mathbb{E}^{\mathbb{Q}}[e^{-rh}v(t+h, S_{t+h})]$ to first order in h via Itô forces its h-coefficient to vanish — the Black–Scholes PDE.

Together these are the variational inequality

$$\boxed{\max\left\{g(s) - v,\ v_t + rs\,v_s + \tfrac{1}{2}\sigma^2 s^2 v_{ss} - rv\right\} = 0.}$$

The terminal condition is $v(T,s) = g(s)$.

WATCH OUT Unlike a European option, the region where the PDE holds is not known in advance. Finding it is part of the pricing problem.

The put PDE needs terminal and spatial conditions

Assume no dividends, $r \geq 0$, and $g(s) = (K - s)^+$. Write

$$\mathcal{A}v = rs\, v_s + \tfrac{1}{2}\sigma^2 s^2 v_{ss} - rv.$$

The obstacle problem can be written as three conditions:

$$\boxed{v - g \geq 0, \qquad -(v_t + \mathcal{A}v) \geq 0, \qquad (v - g)(v_t + \mathcal{A}v) = 0.}$$

Data for the backward calculation

$$v(T, s) = (K - s)^+, \qquad v(t, 0) = K, \qquad \lim_{s \to \infty} v(t, s) = 0.$$

On a finite grid, choose $S_{\max} > K$ and approximate $v(t, S_{\max}) = 0$. Check the effect of increasing $S_{\max}$.

Continue: $v > g$ forces the PDE equality. **Exercise:** $v = g$.

For an American put, the boundary is the decision rule

For a put, there is typically a boundary $s^*(t)$ such that

$$\begin{cases} \text{exercise}, & s \le s^*(t), \\ \text{continue}, & s > s^*(t). \end{cases}$$

At the boundary, the value joins the payoff smoothly:

$$\boxed{v(t, s^*(t)) = K - s^*(t), \qquad v_s(t, s^*(t)) = -1.}$$

Boundary conditions

At maturity, $v(T, s) = (K - s)^+$. For positive rates, the put boundary typically approaches K from below as maturity approaches.

Why a non-dividend-paying American call is European

Let $C^{\mathrm{Eu}}(t,s)$ be a European call and suppose $r \geq 0$, with no dividends. For every $t < T$,

$$C^{\mathrm{Eu}}(t,s) \geq s - Ke^{-r(T-t)} \geq s - K.$$

Thus the European call already dominates the value from immediate exercise, $(s-K)^+$.

NO-ARBITRAGE Waiting until maturity is feasible for the American holder, and immediate exercise never improves on the European value. Therefore

$$\boxed{C^{\mathrm{Am}} = C^{\mathrm{Eu}} \quad \text{(no dividends)}.}$$

Dividends change the call trade-off

No dividend

The option holder does not receive a stock dividend. Giving up the option to buy the stock early has no compensating cash-flow benefit.

Known dividend

Exercising before an ex-dividend date can give ownership of the stock and therefore the dividend. Early exercise of a call can become optimal.

YOUR TURN Which object changes with dividends: the payoff, stopping-time logic, or the continuation value?

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Answer

The payoff and stopping-time logic remain. Dividends alter the economics of waiting, hence the continuation value and the exercise boundary.

Computing American prices

A binomial tree is backward induction made numerical

With N steps of length $\Delta t = T/N$, the CRR parameters are

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = u^{-1}, \quad p = \frac{e^{r\Delta t} - d}{u - d}.$$

At every node (n, j), compute

$$\boxed{V_n^j = \max\left\{g(S_n^j), e^{-r\Delta t}\left[pV_{n+1}^{j+1} + (1-p)V_{n+1}^j\right]\right\}.}$$

Output

The tree supplies both a price and a discrete approximation to the exercise boundary: mark every node at which the maximum selects $g(S_n^j)$.

Finite differences: build the grid and start at maturity

For the no-dividend put, with $r \geq 0$, set

$$s_i = i\Delta s, \quad \Delta s = S_{\max}/M, \qquad t_n = n\Delta t, \quad \Delta t = T/N, \qquad V_i^n \approx v(t_n, s_i).$$

Start with $V_i^N = g_i = (K - s_i)^+$ and work **backward**: $n = N-1, \ldots, 0$.

Centered differences at an interior stock node

$$v_s(t_n, s_i) \approx \frac{V_{i+1}^n - V_{i-1}^n}{2\Delta s}, \qquad v_{ss}(t_n, s_i) \approx \frac{V_{i+1}^n - 2V_i^n + V_{i-1}^n}{(\Delta s)^2}.$$

Boundary values and the obstacle

At every time level impose $V_0^n = K$, $V_M^n = 0$. The obstacle is $V_i^n \geq g_i$; the price is read at $t_0 = 0$.

Explicit scheme: a continuation step, then the maximum

Discretizing $v_t + \mathcal{A}v = 0$ with spatial terms at the known level $n+1$ gives, for $i = 1, \dots, M-1$,

$$\boxed{C_i^n = a_i V_{i-1}^{n+1} + b_i V_i^{n+1} + c_i V_{i+1}^{n+1}, \qquad V_i^n = \max\{g_i, C_i^n\}.}$$

Here $s_i = i\Delta s$, so

$$a_i = \tfrac{\Delta t}{2}(\sigma^2 i^2 - ri), \qquad c_i = \tfrac{\Delta t}{2}(\sigma^2 i^2 + ri),$$
$$b_i = 1 - \Delta t(\sigma^2 i^2 + r), \quad a_i + b_i + c_i = 1 - r\Delta t.$$

Check the weights before using this explicit scheme

Require $a_i, b_i, c_i \geq 0$ at every interior node. For this centered stencil:

$$\sigma^2 i^2 \geq ri, \qquad \Delta t \leq \frac{1}{\sigma^2(M-1)^2 + r}.$$

If a spatial weight is negative, change the drift stencil; reducing Δt alone cannot fix its sign.

One time step: exercise at 80, continue at 100

Take $K = 100$, $r = 5\%$, $\sigma = 30\%$, $\Delta s = 5$, $\Delta t = 1/252$, and $S_{\max} = 200$. At the last step, the known values are $V_i^N = (100 - s_i)^+$.

s_i	$(V_{i-1}^N, V_i^N, V_{i+1}^N)$	C_i^{N-1}	g_i	Decision
80	$(25, 20, 15)$	19.9802	20	Exercise
100	$(5, 0, 0)$	0.3472	0	Continue

The PDE step and the stopping decision are both visible

At $s = 80$, $C = 20 - rK\Delta t < 20$: the maximum restores 20. At $s = 100$, $C = 5a_{20} > 0$: keeping the option has value. These are grid approximations one time step before maturity.

Mark nodes where $V_i^n = g_i > 0$ to approximate the put exercise region.

Implicit scheme: solve the PDE and obstacle together

Using spatial terms at the unknown level n, define the residual

$$R_i^n = d_i V_i^n - a_i V_{i-1}^n - c_i V_{i+1}^n - V_i^{n+1}, \qquad d_i = 1 + \Delta t(\sigma^2 i^2 + r).$$

With the same boundary values and a_i, c_i as before, solve

$$\boxed{V_i^n - g_i \geq 0, \quad R_i^n \geq 0, \quad (V_i^n - g_i)R_i^n = 0.}$$

A discrete complementarity problem

Use projected Gauss–Seidel, projected SOR, or policy iteration until the constraints hold. With $a_i, c_i \geq 0$, the implicit scheme avoids the explicit time-step restriction; accuracy still requires refinement.

WATCH OUT One unconstrained implicit solve followed by a single maximum is generally only a splitting approximation, not the solution of this coupled system.

Why plain Monte Carlo is not enough

For a European option, simulation estimates

$$e^{-rT}\mathbb{E}^{\mathbb{Q}}[g(S_T)]$$

by averaging terminal discounted payoffs.

The American difficulty

At an intermediate state, the relevant object is

$$\mathbb{E}^{\mathbb{Q}}[\text{future cash flow} \mid S_t = s],$$

the conditional continuation value. Simulated paths do not reveal this function directly at a single node.

NO-ARBITRAGE American pricing is not just simulation; it is simulation plus a way to estimate the decision rule.

Longstaff–Schwartz learns the continuation value

Least-squares Monte Carlo works backward through simulated paths.

1. Simulate risk-neutral stock paths and start from terminal payoff.
2. At each earlier date, retain in-the-money paths.
3. Regress discounted future cash flows on basis functions, e.g. $1, S_t, S_t^2$.
4. Compare estimated continuation with immediate exercise and update the cash-flow date path by path.
5. Discount the resulting cash flows and average them.

Interpretation

Regression does not price the option directly; it estimates the continuation rule needed to decide whether each path should stop.

Method choice follows the state space

Method	Best fit	Main caution
Binomial tree	One underlying; teaching and robust benchmarks	Many steps may be needed for convergence
Finite differences	Low-dimensional Markov models; boundary analysis	Grid size grows quickly with dimension
Longstaff–Schwartz	Several risk factors or path simulation available	Quality depends on paths, basis, and regression design

NO-ARBITRAGE All three methods implement the same comparison: exercise value versus risk-neutral continuation value.

Three checks catch many implementation mistakes

Before trusting a computed American price, test the output against consequences of the theory.

Check	What should be true
Exercise right	$V^{\mathrm{Am}} \geq V^{\mathrm{Eu}}$ for the same contract.
No-dividend call	$C^{\mathrm{Am}} = C^{\mathrm{Eu}}$, up to numerical tolerance.
Refinement	A tree or grid should stabilize as N increases or the mesh is refined.

YOUR TURN If numerical output gives $V^{\mathrm{Am}} < V^{\mathrm{Eu}}$, is that an economic result or a bug?

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Debug it

Exercise at maturity is feasible, so an American value cannot be lower.

Takeaways

The whole lecture in one recursion

$$V_t = \max\left\{ \underbrace{g(S_t)}_{\text{exercise}}, \underbrace{\beta \mathbb{E}^{\mathbb{Q}}[V_{t+1} \mid \mathcal{F}_t]}_{\text{continue}} \right\}.$$

1. An American option embeds a stopping right, so its value is at least the corresponding European value.
2. The decision is dynamic: optimal exercise is a region of states.
3. In continuous time, that region becomes a free boundary and the PDE becomes an obstacle problem.
4. Trees, PDE solvers, and LSMC differ computationally, not conceptually.

Final check: read the economics before the formula

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Answer

Compare intrinsic value $K - S_t$ with discounted risk-neutral continuation value. The clues make early exercise plausible, but the Bellman comparison makes the decision.

Next: using these tools to value and hedge exercise features.